sEKV Parameter Extraction for the Generic 180nm Process

Long-channel pMOS (Version 1)

02.10.2026

In this notebook we extract the sEKV parameters for a long-channel pMOS transistor from the data generated with the EKV 2.6 compact model for a generic 180nm CMOS process.

Table of contents

1 Introduction **4**

2 Transistor geometry parameters **5**
2.1 Effective length and width for current 5

3 DC Transfer Characteristic Parameters **6**
3.1 Generating and importing the data 6
3.1.1 I_D and G_m versus V_G 6
3.1.2 G_m-V_G and G_m-I_D 7
3.2 Filtering the outliers 7
3.3 Direct extraction with $\lambda_c = 0$ 10
3.3.1 Slope factor n and I_{spec} extraction 10
3.3.2 Threshold voltage extraction 12
3.3.3 Summary 15
3.4 Extraction using curve fitting with $\lambda_c = 0$ 17
3.4.1 Slope factor n and I_{spec} extraction 17
3.5 Comparison with parameters extracted from the EKV 2.6 compact model 17

4 Output characteristic **20**
4.1 Generating and importing the data 20
4.1.1 I_D and G_{ds} versus V_D 20
4.1.2 Filtering the outliers 21
4.2 Extracting the CLM parameter 21

5 Noise **23**
5.1 Setting the bias conditions 23
5.2 Extract operating point information 23
5.3 Input-referred noise PSD 24

6 Conclusion **26**

References **27**

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1 Introduction

In this notebook we will extract the sEKV parameters [1] [2] for a long-channel pMOS transistors from a generic 180nm bulk CMOS process. The physical parameters are given in Table 1.1, the global process parameters in Table 1.2.

Table 1.1: Physical parameters

Parameter	Value	Unit
T	300	K
U_T	25.875	mV

Table 1.2: Global process parameters

Parameter	Value	Unit
V_{DD}	1.8	V
C_{ox}	8.46	$\frac{fF}{\mu m^2}$
W_{min}	200	nm
L_{min}	180	nm

The easiest way to extract the sEKV parameters is to use the python tool developed by H.C. Han and available on GitLab [3]. The tool and the extraction procedure are described in [4]. In this notebook we will detail the extraction procedure and show how the parameters can be extracted manually.

We start by looking at the channel width and length corrections for the drain current and for the capacitances. Then we will extract the sEKV parameter using a direct extraction methodology with the velocity parameter $\lambda_c = 0$. We then will extract the additional parameter λ_c using a direct extraction methodology. Then we will extract all the sEKV parameters by optimization using nonlinear curve fitting.

We also will extract the output conductance due to channel-length modulation (CLM) and the related parameter.

Finally, we will check the white noise model and extract the EKV flicker noise parameters.

The extracted parameters are then all saved in an Excel worksheet.

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2 Transistor geometry parameters

2.1 Effective length and width for current

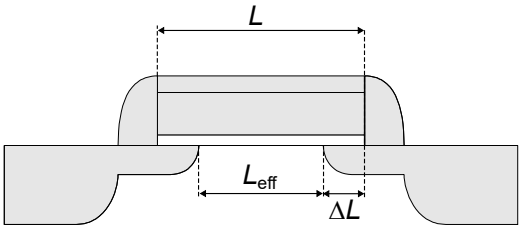


Figure 2.1: Definition of the effective length.

The effective length and width in the EKV 2.6 compact model are defined as [5]

$$W_{eff} = W + \Delta W, \tag{2.1}$$
$$L_{eff} = L + \Delta L, \tag{2.2}$$

where the width and length correction parameters ΔW and ΔL are given in Table 2.1.

Table 2.1: Transistor geometry.

Parameter	Symbol	Value	Unit
Selected transistor width and length			
Width	W	10	μm
Length	L	10	μm
Width and length correction parameters			
Width correction	ΔW	54	nm
Length correction	ΔL	-72	nm
Selected transistor effective width and length			
Effective width	W_{eff}	10.054	μm
Effective length	L_{eff}	9.928	μm

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3 DC Transfer Characteristic Parameters

3.1 Generating and importing the data

The data used for the sEKV parameters extraction is generated by simulation with ngspice using the EKV 2.6 compact model [5] for a generic 180nm bulk CMOS process. We present the I_D-V_G and G_m-V_G data below.

3.1.1 I_D and G_m versus V_G

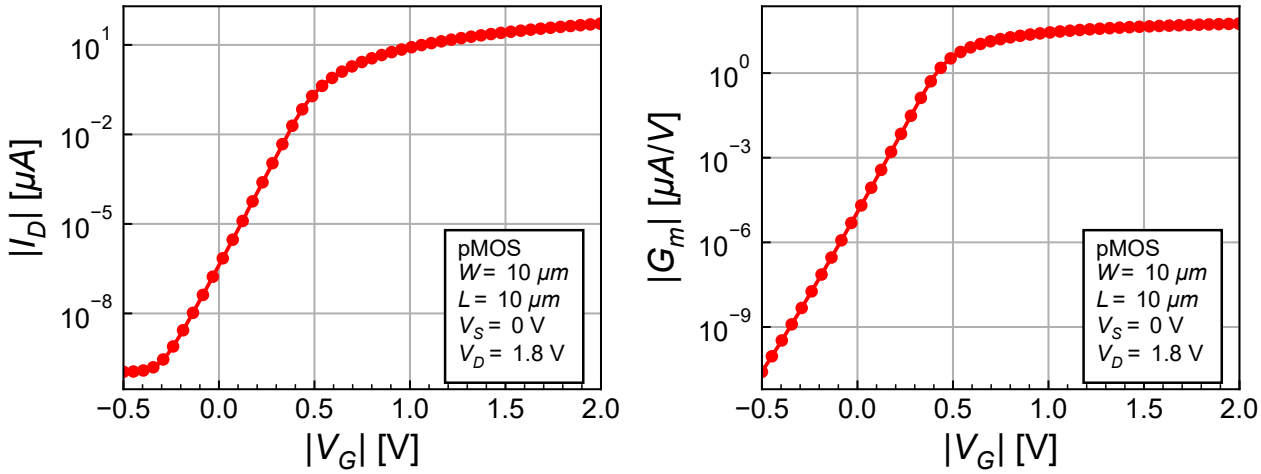


Figure 3.1: Imported I_D-V_G and G_m-V_G.

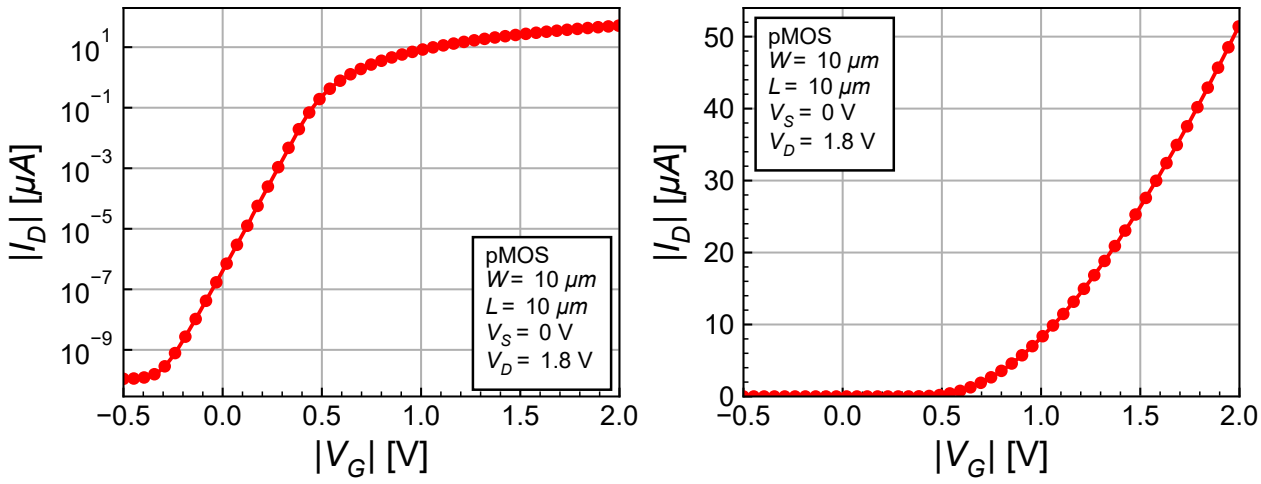


Figure 3.2: Imported I_D-V_G.

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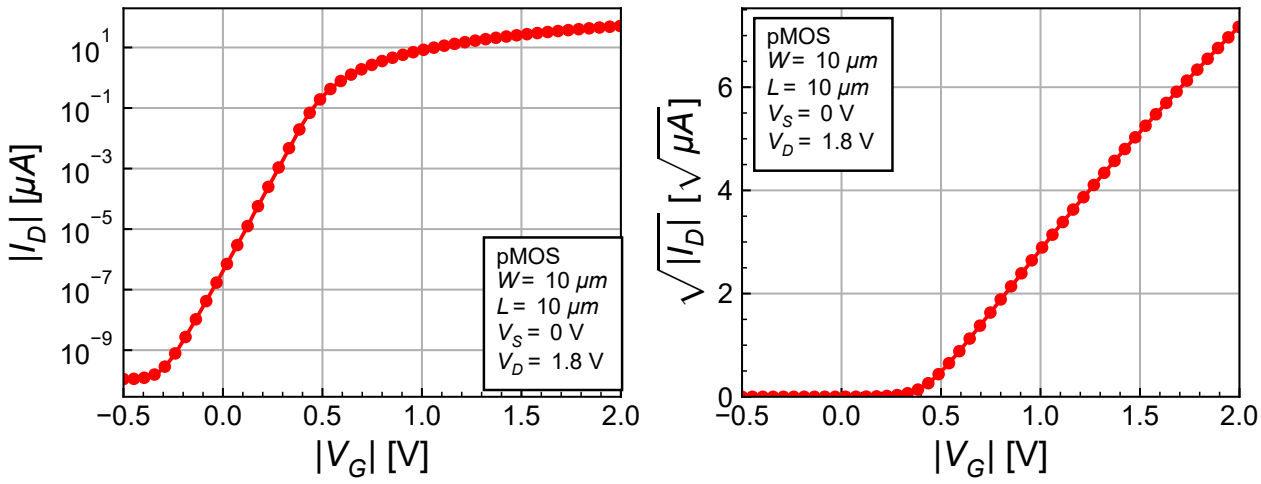


Figure 3.3: Imported I_D-V_G.

3.1.2 G_m-V_G and G_m-I_D

We now will check the derivative namely the gate transconductance obtained from the simulator and compare it to the numerical differentiation of the large-signal I_D-V_G characteristic.

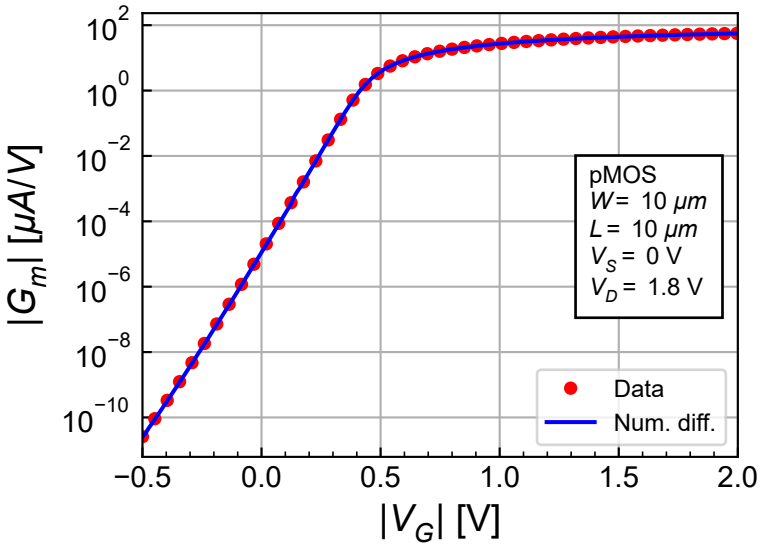


Figure 3.4: Check of the G_m-V_G consistency.

We see that the transconductance obtained by differentiating the large-signal I_D-V_G characteristic is equal to the transconductance extracted from the EKV 2.6 compact model. We will keep the value extracted from the EKV 2.6 compact model.

3.2 Filtering the outliers

Since the sEKV model [2] doesn't account for leakage currents such as gate induced drain leakage (GIDL) that appears at very low current, we need to filter the outlier points. This is done below with the outlier points shown in blue and the points used for the extraction shown in red.

We can now proceed with the parameter extraction, starting with the direct extraction approach.

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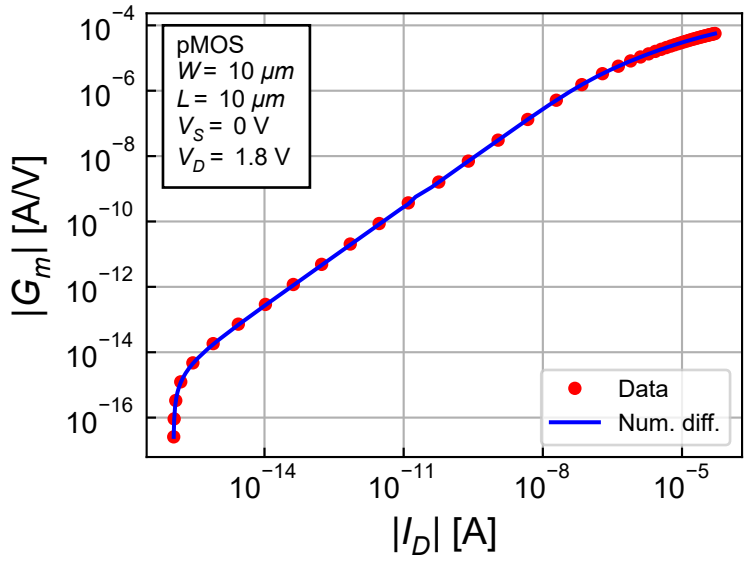


Figure 3.5: Check of the G_m-I_D consistency.

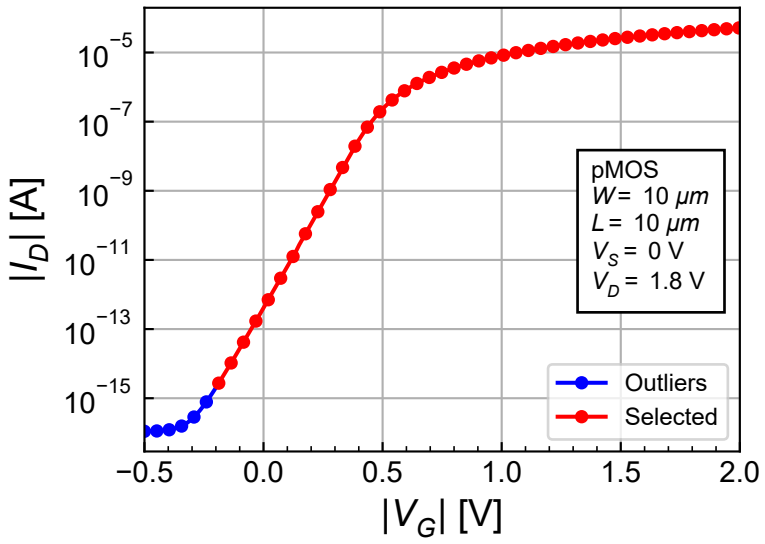


Figure 3.6: Filtering the outliers from the I_D-V_G charactersitic.

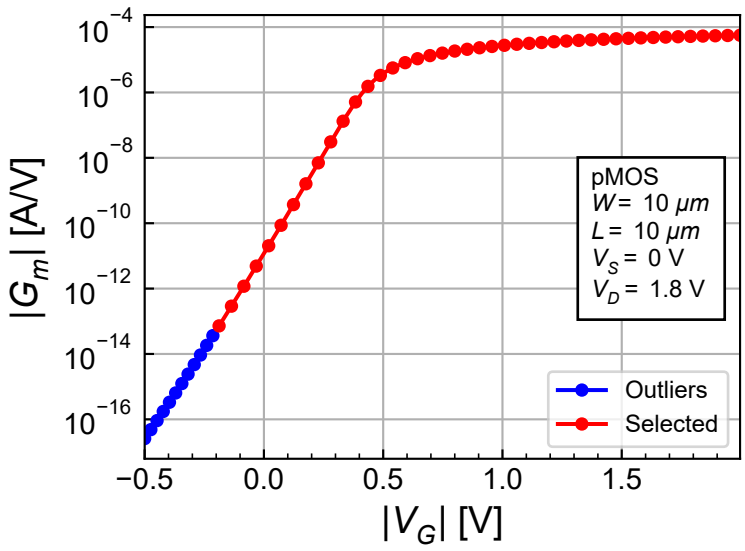


Figure 3.7: Filtering the outliers from the G_m-V_G characersitic.

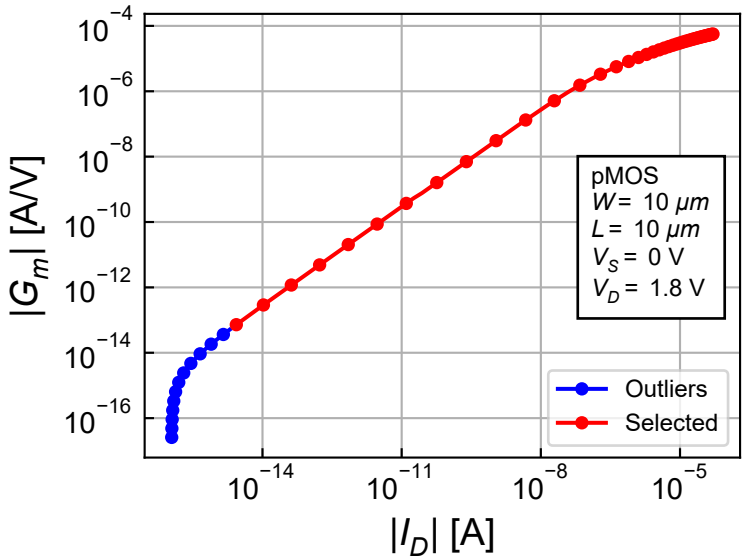


Figure 3.8: Filtering the outliers from the G_m-I_D characersitic.

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3.3 Direct extraction with $\lambda_c = 0$

In the direct extraction approach, we avoid using curve fitting or optimization and manipulate the data to extract a given parameter in a certain data range.

3.3.1 Slope factor n and I_{spec} extraction

The gate transconductance in weak inversion and saturation is given by [1]

$$G_m = \frac{I_D}{n\,U_T}. \tag{3.1}$$

So if we plot $I_D/(G_m\,U_T)$ we should see a plateau in weak inversion the value of which is equal to the slope factor n.

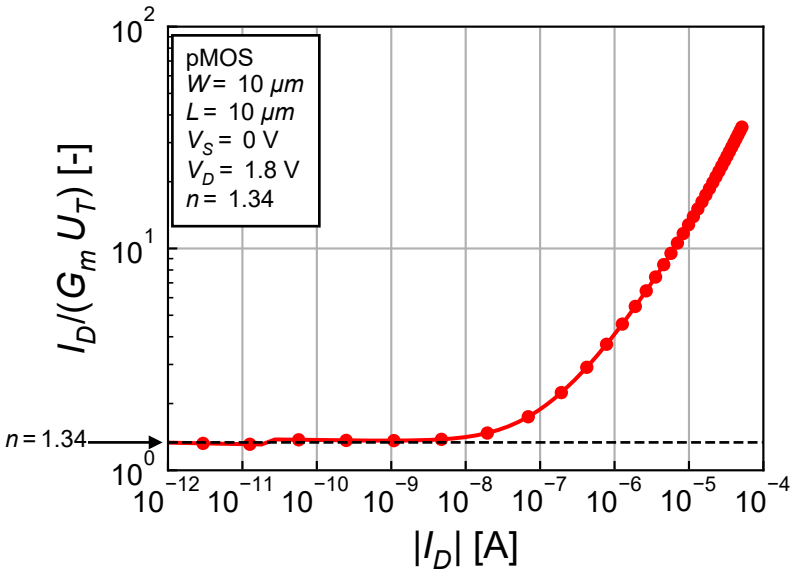


Figure 3.9: Slope factor extraction.

This is illustrated in Figure 3.9 resulting in $n = 1.34$. On the other hand the normalized G_m/I_D function for a long-channel transistor in strong inversion and saturation is given by [1]

$$\frac{G_m\,n\,U_T}{I_D} = \frac{1}{\sqrt{IC}} = \sqrt{\frac{I_{spec}}{I_D}}. \tag{3.2}$$

We can then plot $(G_m\,n\,U_T)^2/I_D$ which should find a maximum value equal to I_{spec}.

This is illustrated in Figure 3.10 resulting in $I_{spec} = 115\ nA$ corresponding to $I_{spec\square} = 114\ nA$. We can now plot $I_D/(G_m\,U_T)$ versus I_D as shown in Figure 3.11. We clearly see the two asymptotes in weak (i.e. $I_D < I_{spec}$) and strong inversion (i.e. $I_D > I_{spec}$).

Having extracted n and I_{spec}, we can now plot the normalized G_m/I_D function versus IC which is shown in Figure 3.12.

The fit is reasonable over the entire IC span. There is some discrepancy in the strong inversion region which is due to the mobility reduction due to the vertical field appearing for $IC > 10^2$. The latter can be accounted for by using the λ_c parameter which is normally used for modeling the effect of velocity saturation in short-channel transistor but can also be used to correct the effect of mobility reduction due to the vertical field appearing in long-channel transistors. We will not do this here since we want to extract the long-channel parameters keeping $\lambda_c = 0$, but since we are mostly interested in

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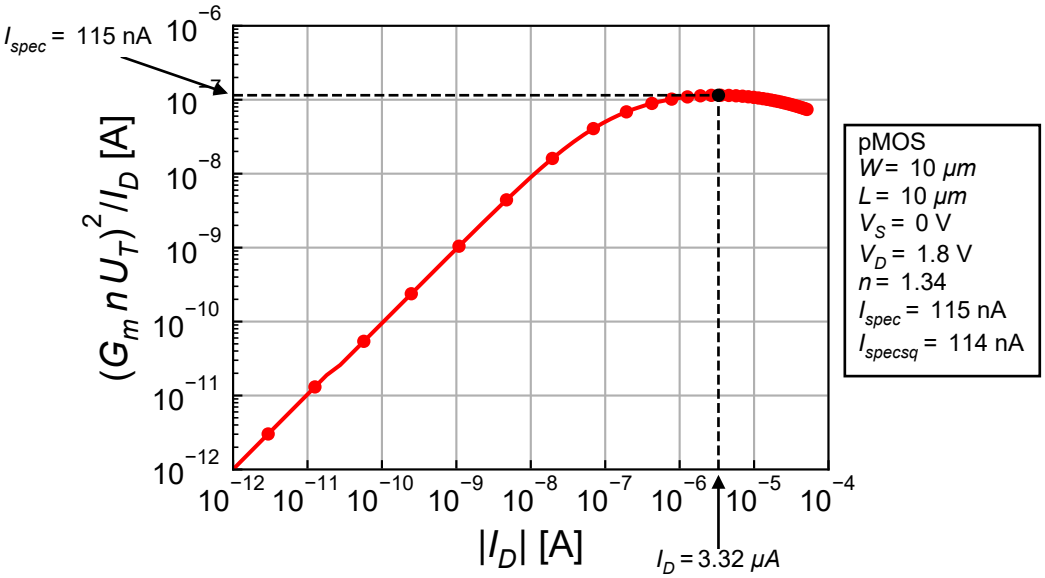


Figure 3.10: I_{spec} extraction.

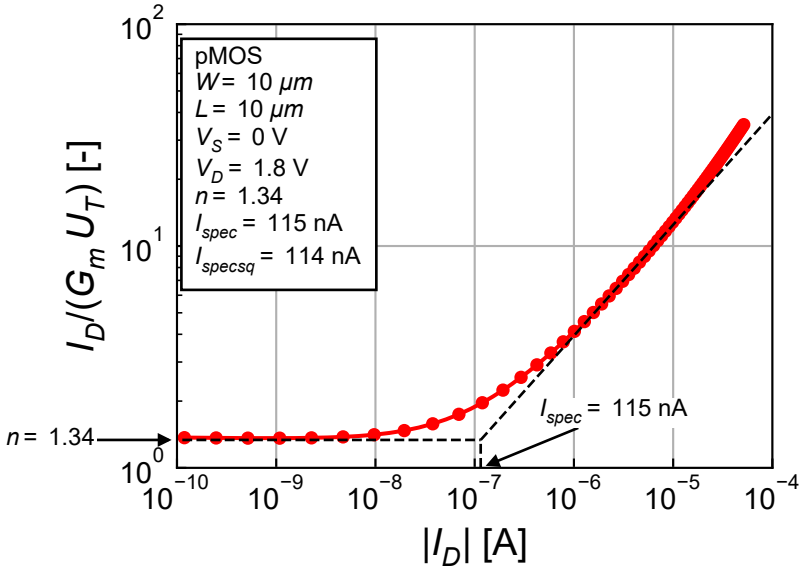


Figure 3.11: n and I_{spec} check.

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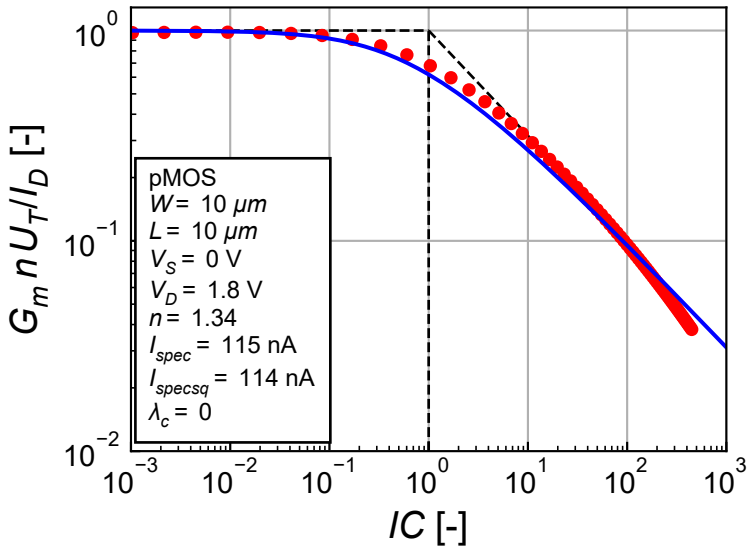


Figure 3.12: Resulting normalized G_m/I_D characteristic.

the moderate inversion region, we can slightly increase I_{spec} to improve the fit in moderate inversion at the cost of a degradation in strong inversion. This results in the normalized G_m/I_D function versus IC shown in Figure 3.13.

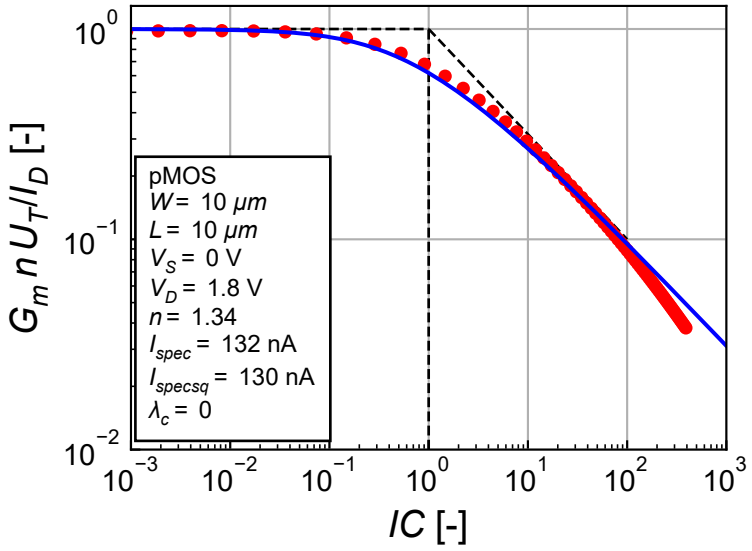


Figure 3.13: Fine tuning of the normalized G_m/I_D function versus IC in moderate inversion.

The fit is now much better in moderate inversion but less in strong inversion. This is due to mobility reduction due to the vertical field which is an effect that is not accounted for in the model. However, it offers a good trade-off between moderate inversion (where the curve is slightly below the simulations) and strong inversion up to $IC = 100$ (where the simulation points are slightly below the curve). We therefore will keep the new value of $I_{spec\square}$, namely $I_{spec\square} = 130\ nA$.

3.3.2 Threshold voltage extraction

We can extract the threshold voltage in weak inversion (assuming $V_S = 0$) from the normalized current (inversion coefficient) given by [1]

$$IC = e^{\frac{V_G - V_{T0}}{nU_T}}. \tag{3.3}$$

We can now plot

$$V_{T0} = V_G - nU_T \ln(IC) \tag{3.4}$$

to extract the threshold voltage. This results in the plot shown in Figure 3.14.

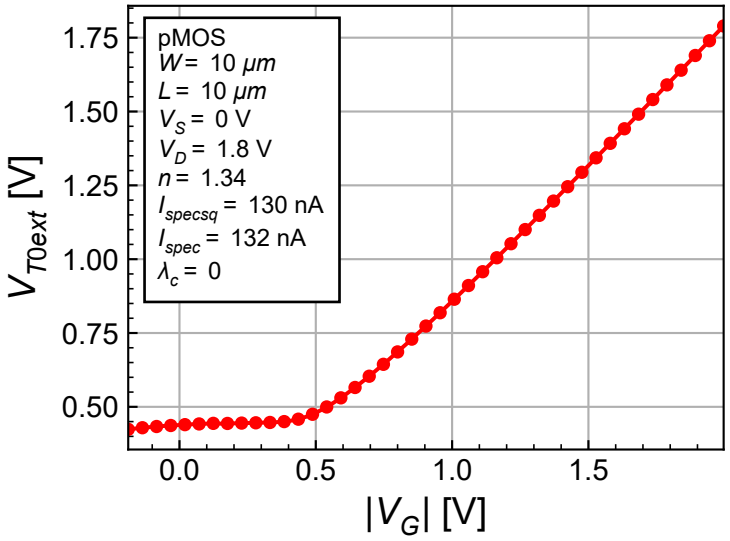


Figure 3.14: Threshold voltage extraction.

We see a plateau in weak inversion where we can average its value to get the threshold voltage in weak inversion which we can zoom into as shown in Figure 3.15.

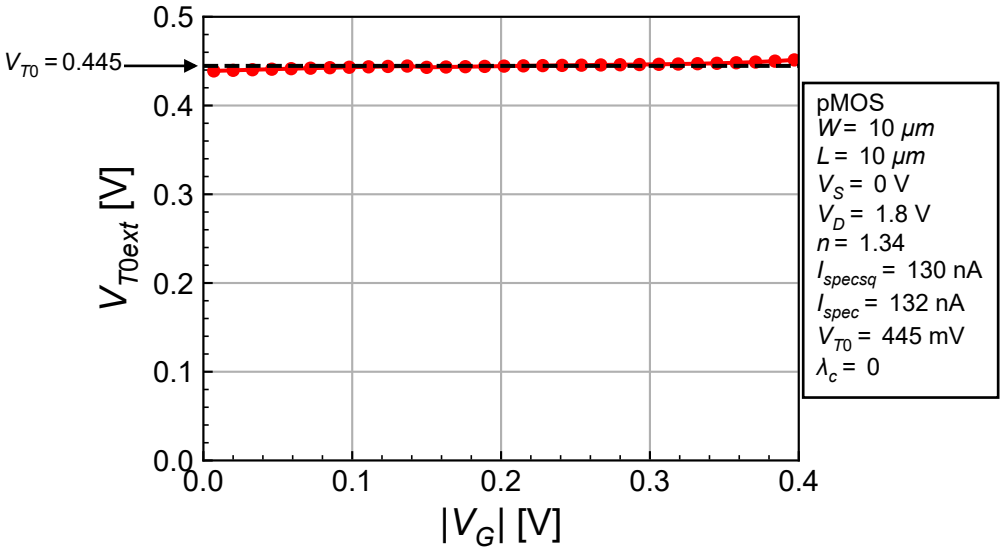


Figure 3.15: Threshold voltage extraction in weak inversion.

The average value of $V_{T0} =$ in this range is given by $V_{T0} = 445\ mV$.

We can now plot the I_D-V_G for the extracted parameters which is shown in Figure 3.16.

The fit is good, but I_{spec} can be slightly reduced.

We now get a good fit with some deviations in strong inversion, which is expected since we focused on the moderate inversion and kept $\lambda_c = 0$. Anyway, when designing circuits we rarely we use an inversion coefficient larger than 100 corresponding for this technology to an overdrive voltage $V_G - V_{T0} \cong 0.7\,V$. The fit within this range is therefore acceptable.

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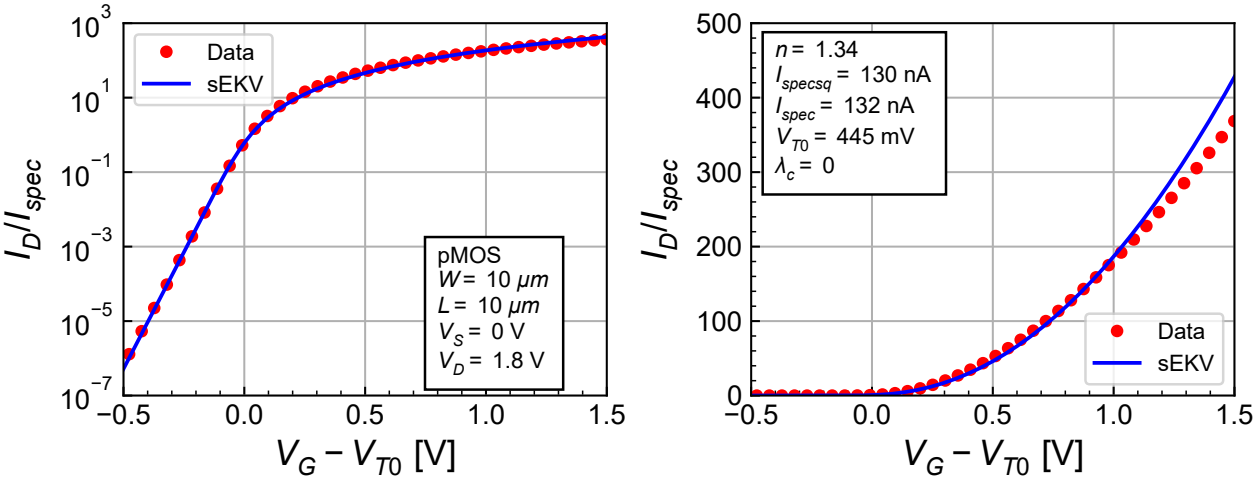


Figure 3.16: I_D-V_G for the extracted parameters.

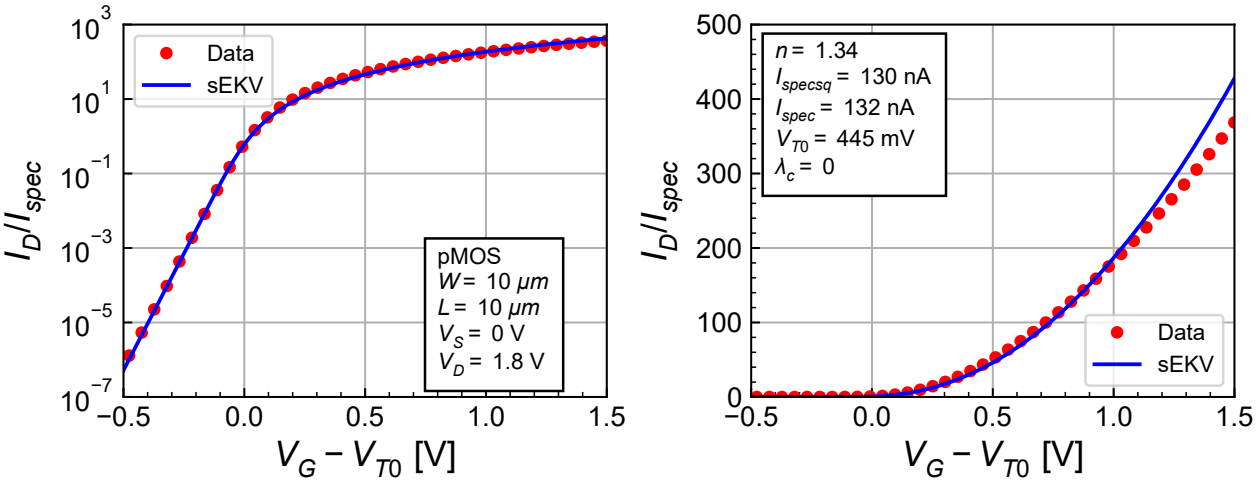


Figure 3.17: I_D-V_G for the extracted parameters.

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We can also extract the threshold voltage in strong inversion by plotting $\sqrt{I_D}$ versus V_G as shown in Figure 3.18.

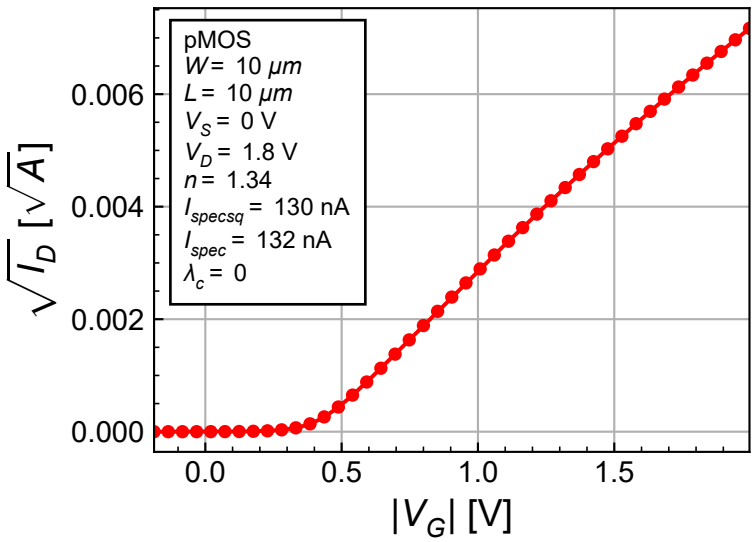


Figure 3.18: $\sqrt{I_D}$-V_G for the extracted parameters.

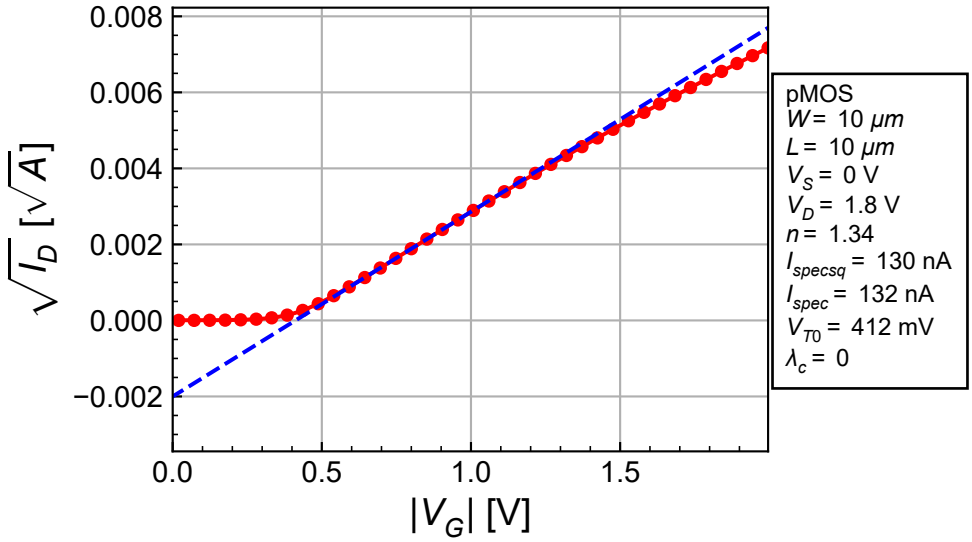


Figure 3.19: Threshold voltage extraction from $\sqrt{I_D}$-V_G in strong inversion.

We can then fit the linear portion and extract the intersection point on the V_G axis as shown in Figure 3.19 which results in $V_{T0} = 412\ mV$. We get a very small threshold voltage even smaller than the value extracted in weak inversion. We can check the I_D-V_G charactersitic with this extracted threshold voltage in Figure 3.20.

As expected, we get a less good fit in weak inversion. We therefore keep the value of the threshold voltage extracted in weak inversion, namely $V_{T0} = 445\ mV$.

3.3.3 Summary

The results of the direct extraction method is shown in Figure 3.21, which includes the large-signal $IC = I_D/I_{spec}$ versus $V_G - V_{T0}$ on the left and the small-signal parameters G_{ms}/G_{spec} versus IC and $G_m\, n\, U_T/I_D$ versus IC on the right. We see a good fit of the large- and small-signal parameters except in very strong inversion (i.e. $100 < IC$). The extracted parameters are summarized in Table 3.1.

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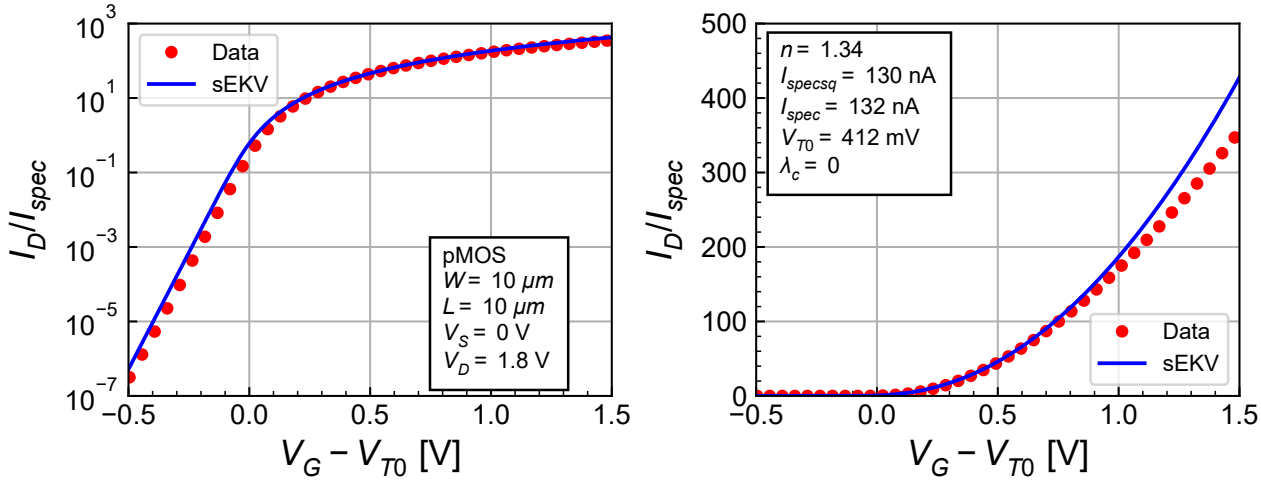


Figure 3.20: I_D-V_G for the extracted threshold voltage.

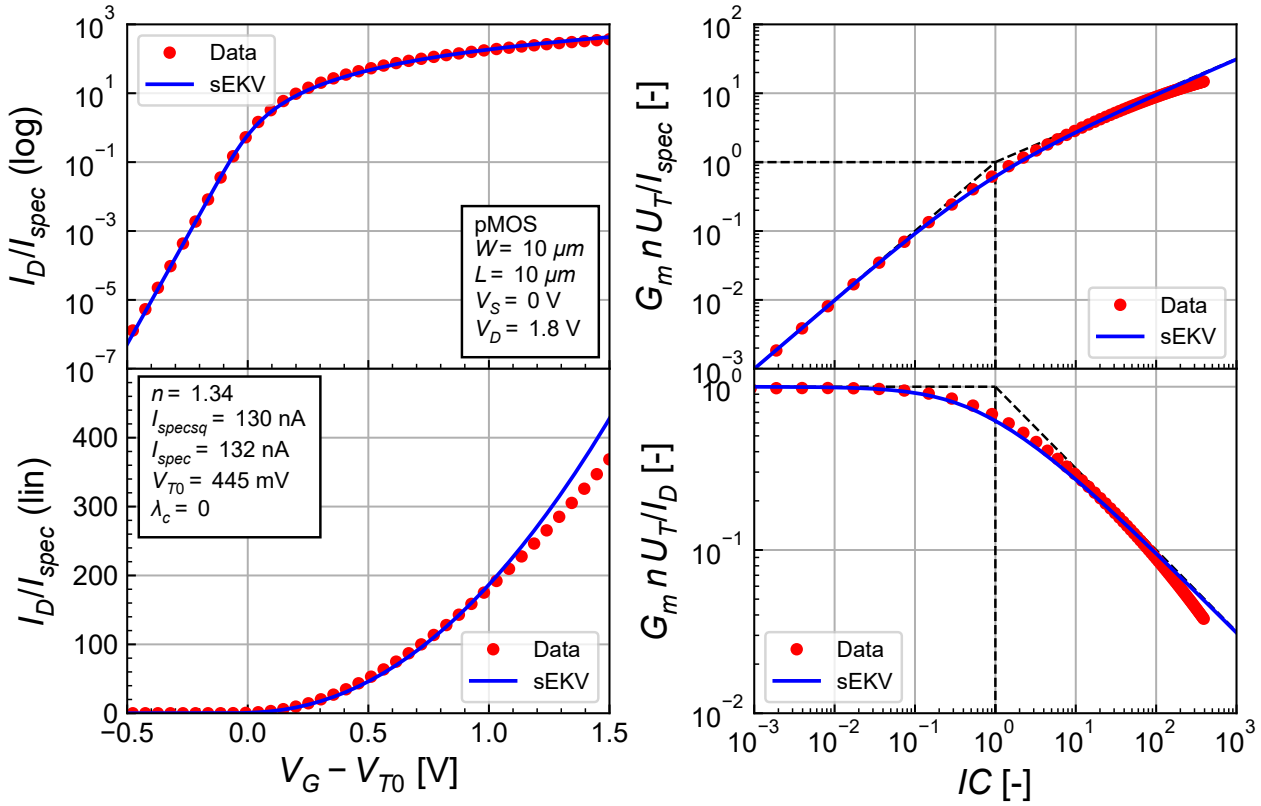


Figure 3.21: Summary of direct extraction with $\lambda_c = 0$.

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Table 3.1: Direct extraction of the sEKV parameters with $\lambda_c = 0$.

Type	n	$I_{spec\square}$ $[nA]$	V_{T0} $[mV]$	λ_c	L_{sat} $[nm]$	Comment
pMOS	1.34	130	445	0	0	direct with $\lambda_c = 0$

3.4 Extraction using curve fitting with $\lambda_c = 0$

3.4.1 Slope factor n and I_{spec} extraction

We can extract I_{spec} and V_{T0} for a long-channel directly from $\ln(I_D)$-V_G data using curve fitting keeping $\lambda_c = 0$ and using the vlaue of the slope factor extracted above. The result is shown in Figure 3.22.

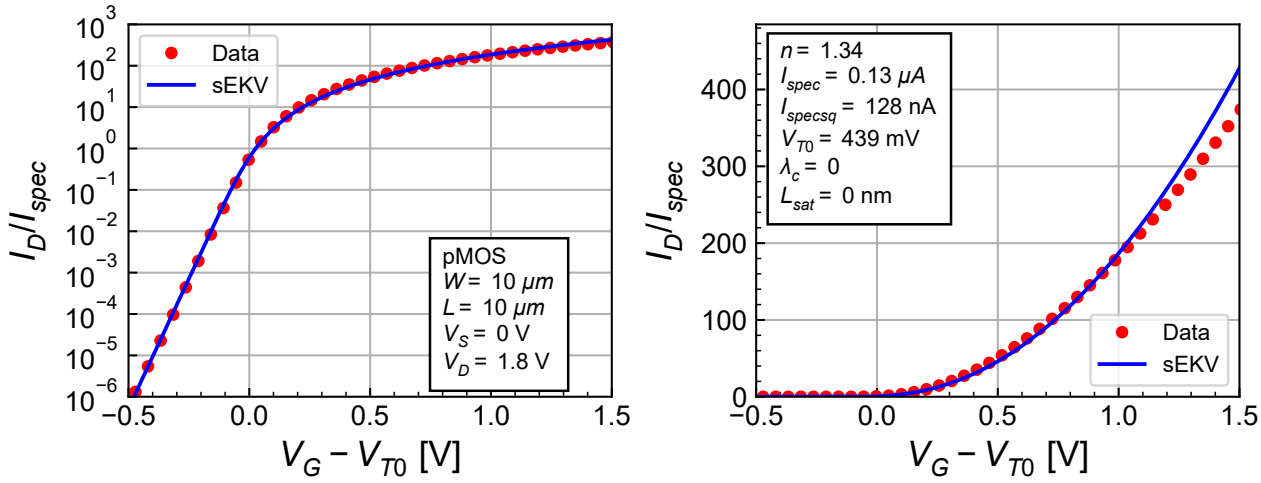


Figure 3.22: Extraction of I_{spec} and V_{T0} by curve fitting.

The fit is very good, remembering that $\lambda_c = 0$.

Table 3.2: Extraction of the sEKV parameters using curve fitting with $\lambda_c = 0$.

Type	n	$I_{spec\square}$ $[nA]$	V_{T0} $[mV]$	λ_c	L_{sat} $[nm]$	Comment
pMOS	1.34	130	445	0	0	direct with $\lambda_c = 0$
pMOS	1.34	128	439	0	0	curve fitting with $\lambda_c = 0$

3.5 Comparison with parameters extracted from the EKV 2.6 compact model

We can compare the data with the parameters taken directly from the EKV 2.6 compact model parameter deck. More precisely, V_{T0} is taken equal to the value used in EKV 2.6, namely $V_{T0} = 0.457$ V. The slope factor is calculated as

$$n = n_0 = 1 + \frac{\Gamma}{2\sqrt{\Psi_0}}, \tag{3.5}$$

where $\Gamma = 0.609\ \sqrt{V}$ and $\Psi_0 = 0.990\ V$. $I_{spec\square}$ is calculated as

$$I_{spec\square} = 2\ n\ \beta\ U_T^2, \tag{3.6}$$

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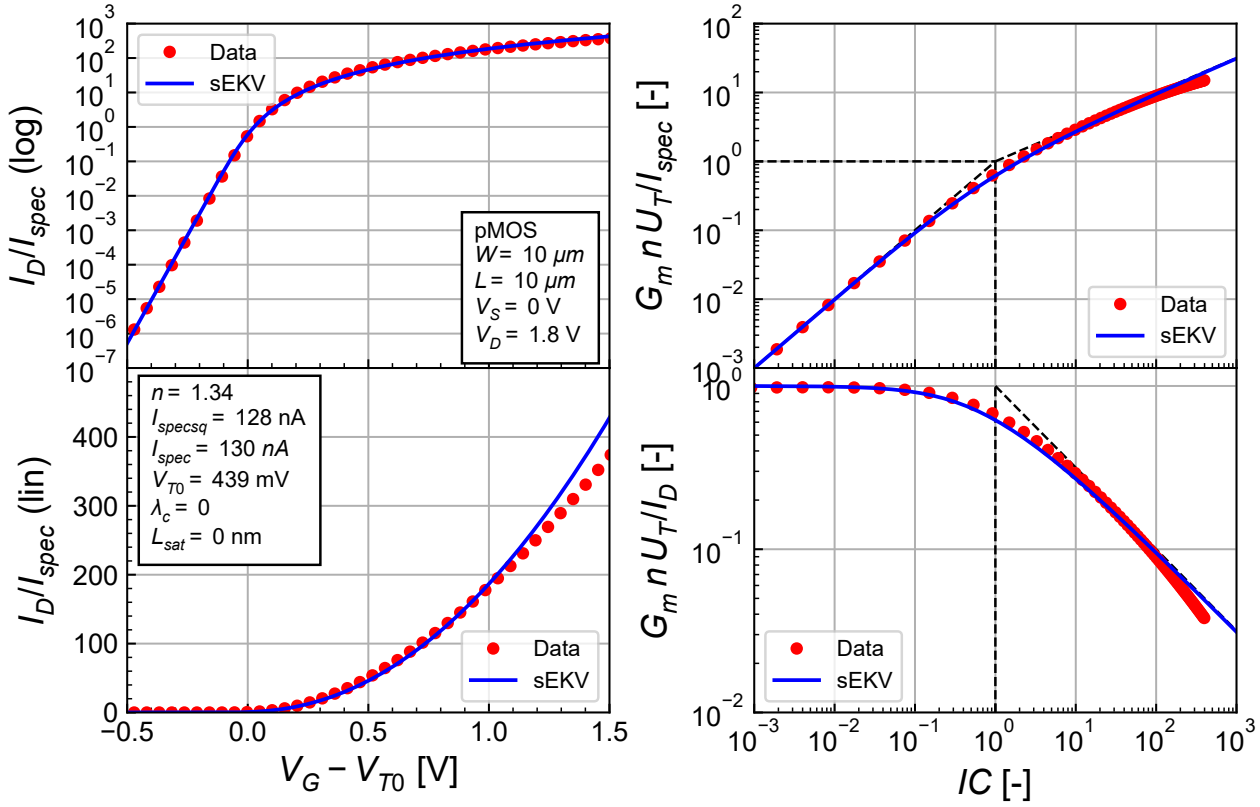


Figure 3.23: Summary of extraction by curve fitting.

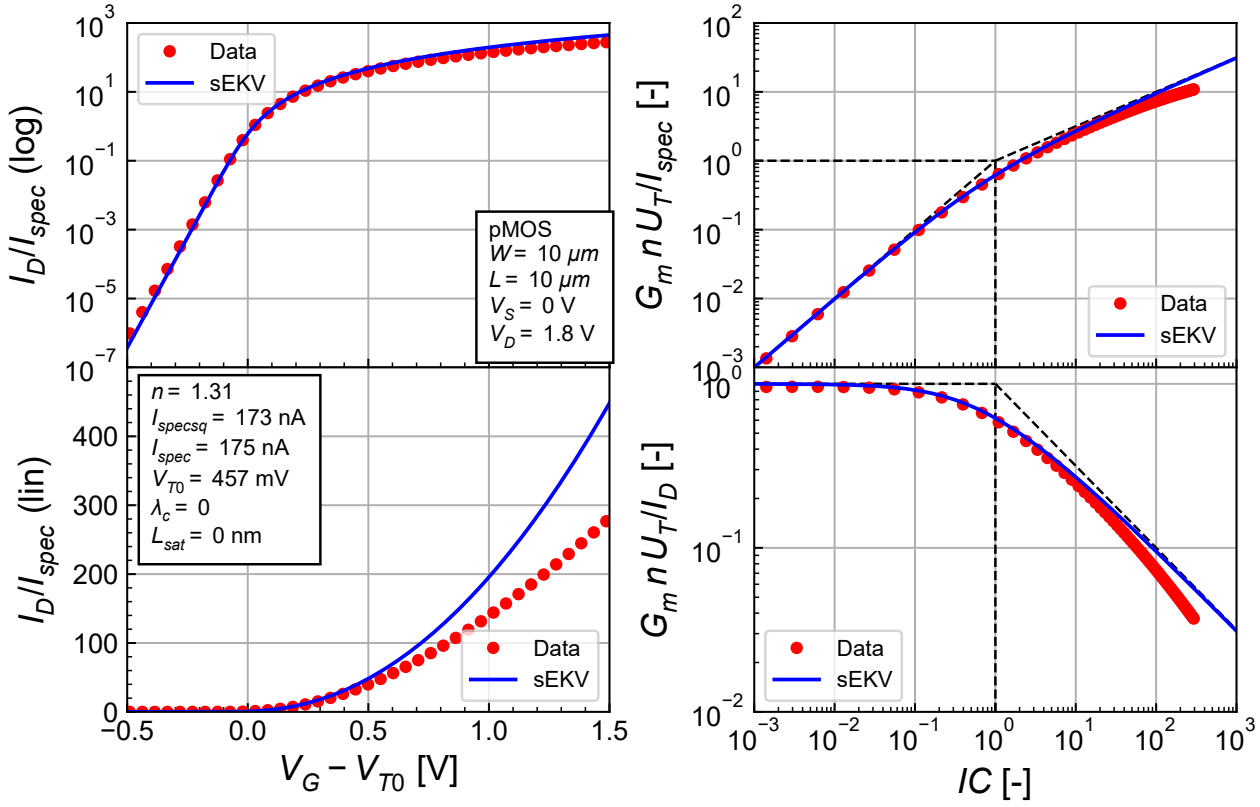


Figure 3.24: Result for the parameters calculated from the EKV 2.6 CM parameters with $\lambda_c = 0$.

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with $\beta = 99\ \mu A/V^2$. The parameter L_{sat} is not included in EKV 2.6. We can set it to $\lambda_c = 0$ resulting in the fit shown in Figure 3.24.

The fit is less accurate than those obtained above.

We can check including the value of $L_{sat} = 12\ nm$. The result is shown in Figure 3.25.

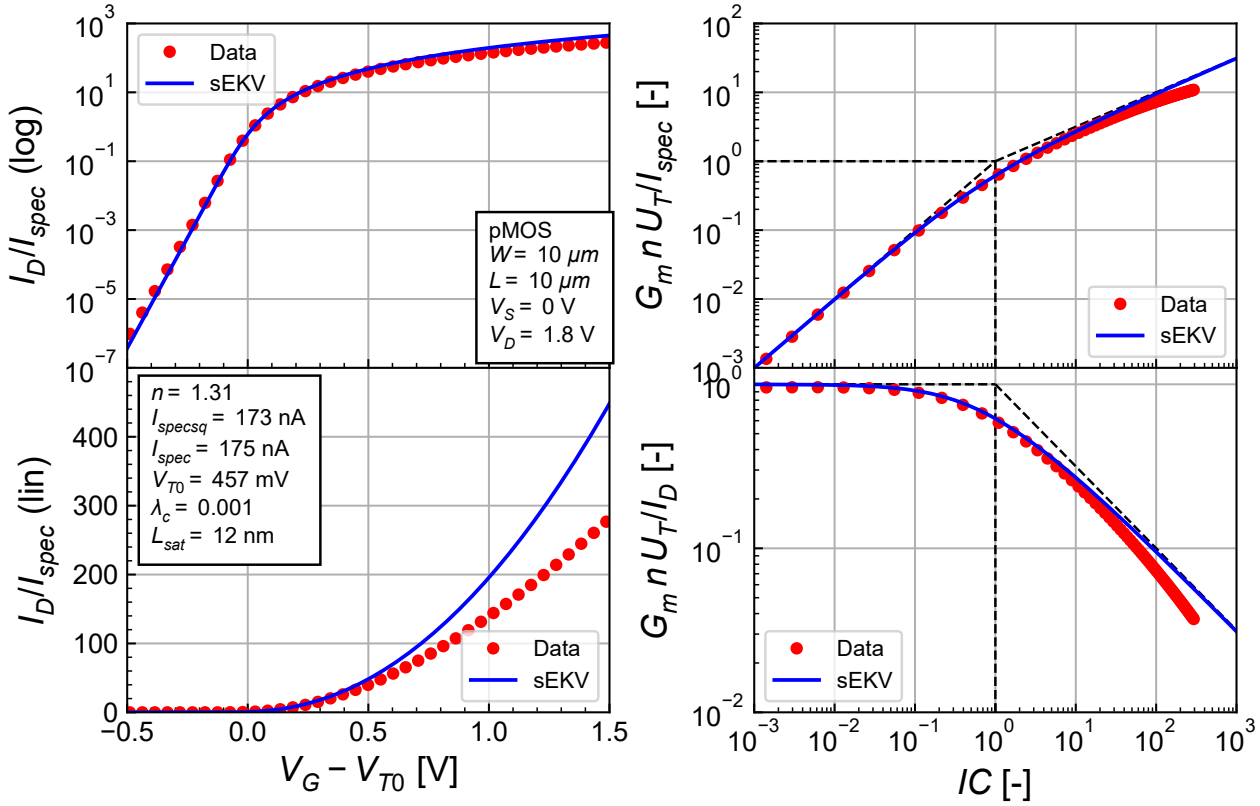


Figure 3.25: Result for the parameters calculated from the EKV 2.6 CM parameters including λ_c.

There is almost no change which is understandable for such a long transistor.

Table 3.3: Extraction of the sEKV parameters using curve fitting with $\lambda_c = 0$.

Type	n	$I_{spec\square}\ [nA]$	$V_{T0}\ [mV]$	λ_c	$L_{sat}\ [nm]$	Comment
pMOS	1.34	130	445	0	0	direct with $\lambda_c = 0$
pMOS	1.34	128	439	0	0	curve fitting with $\lambda_c = 0$
pMOS	1.31	173	457	0	12	From EKV 2.6 parameters

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4 Output characteristic

4.1 Generating and importing the data

In order to extract the CLM parameter in saturation, we will now filter out the points that correspond to the linear region. We can do this easily by looking at the G_{ds} versus V_D characteristic as shown in Figure 4.3. This done below where we will drop the blue points and keep the red points for the extraction.

4.1.1 I_D and G_{ds} versus V_D

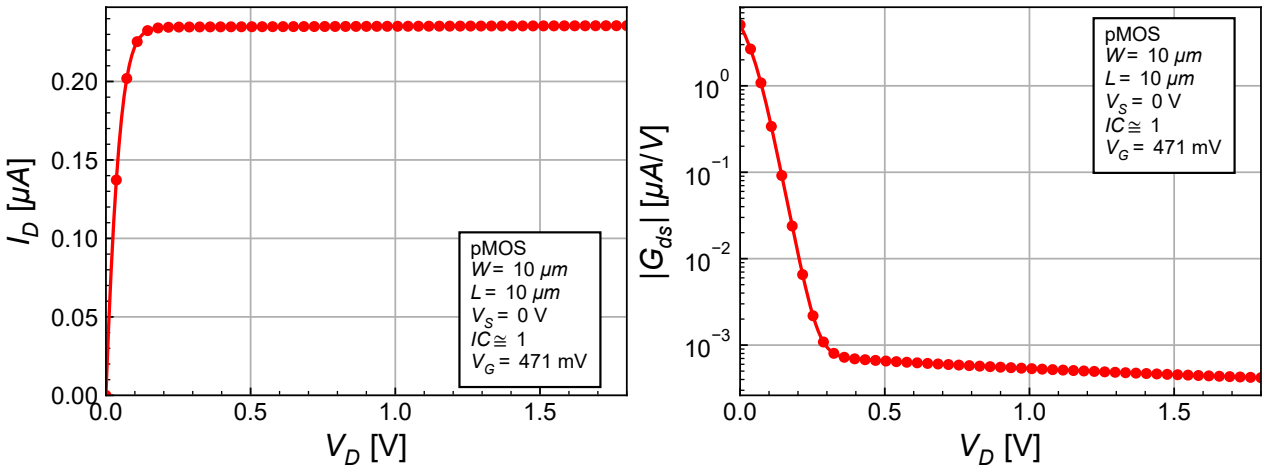


Figure 4.1: Imported I_D-V_D and G_{ds}-V_D.

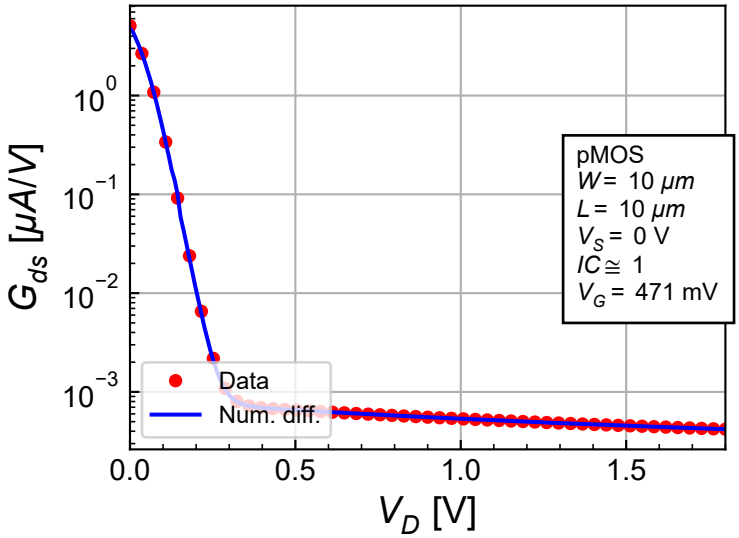


Figure 4.2: Check imported G_{ds} against computed G_{ds}.

The output conductance calculated by differentiating the large-signal I_D-V_D matches the value extracted from the PSP model. We will keep the value from PSP.

4.1.2 Filtering the outliers

In order to extract the CLM parameter in saturation, we will now filter out the points that correspond to the linear region. We can do this easily by looking at the G_{ds} versus V_D characteristic as shown in Figure 4.3.

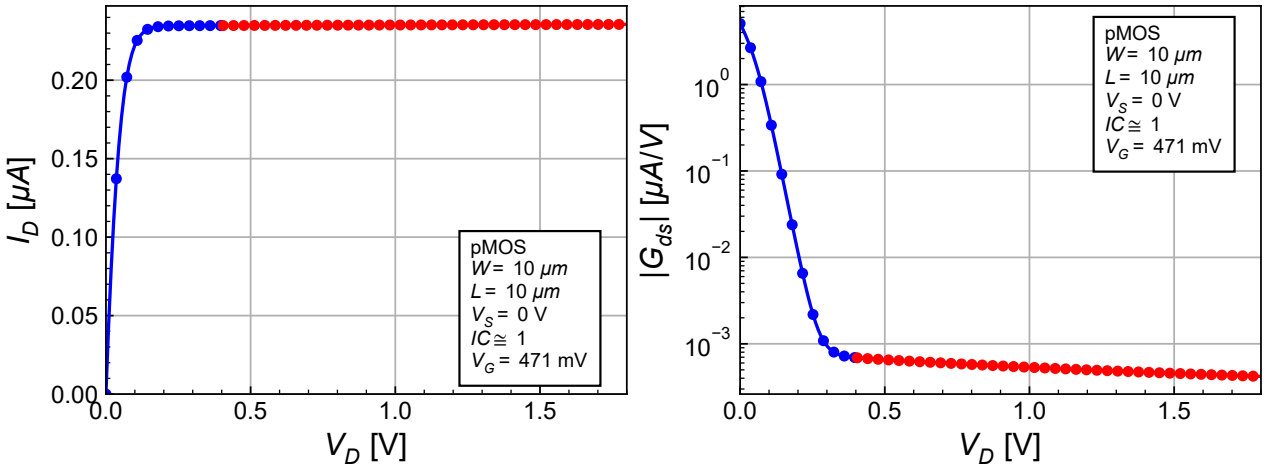


Figure 4.3: Filtering the outliers from I_D-V_G and G_{ds}-V_D.

4.2 Extracting the CLM parameter

The simple channel length modulation (CLM) of the output conductance in saturation is approximated by

$$G_{ds} = \frac{I_{D,sat}}{V_E} \tag{4.1}$$

where $I_{D,sat}$ is the drain current in saturation and

$$V_E = \lambda \cdot L_{eff} \tag{4.2}$$

is the Early voltage or CLM voltage which is proportionnal to the effectve length. This corresponds to a linear approximation of the drain current in saturation given by

$$I_{D,sat} \cong G_{ds} \cdot (V_D - V_E) \tag{4.3}$$

The value I_{D0} corresponds to the intercept

$$I_{D0} = -G_{ds} \cdot V_E \tag{4.4}$$

which is positive since V_E is negative.

As shown in the right plot of Figure 4.4, the output conductance is approximated by a constant value represented by the dashed black line. It is a poor approximation of the output conductance but good enough for starting a design. Note that, in analog circuit design, having a good fit of the output conductance is less important than having a good fit of the transconductance because often we just need the intrinsic voltage gain to be large enough for the circuit to perform correctly, while we want to have a good estimation of the parameters depending on the transconductance, such as the gain-bandwidth product or the thermal noise.

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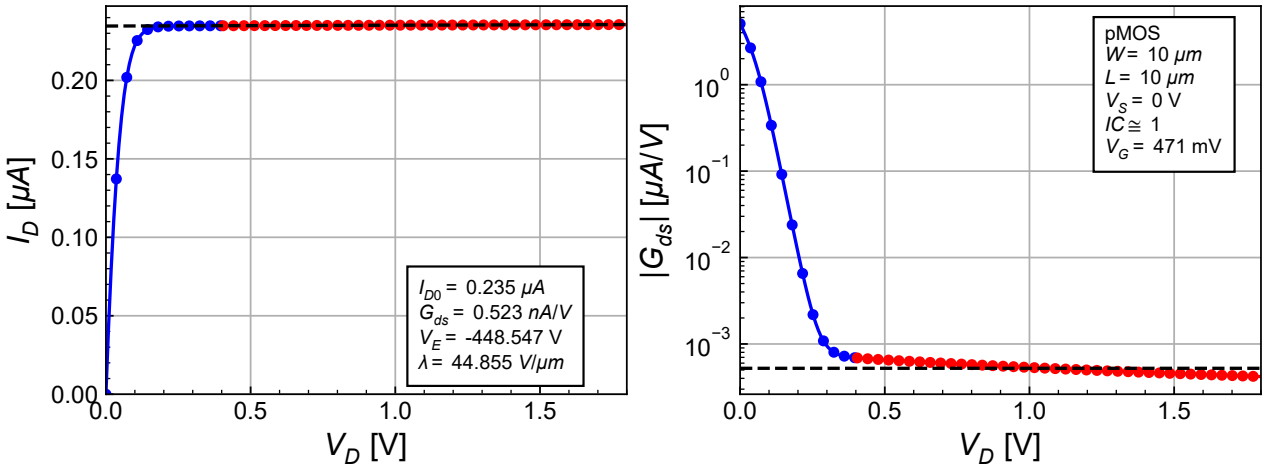


Figure 4.4: Extraction of the CLM parameter λ.

Table 4.1: CLM parameters extracted in moderate inversion.

Type	L_{eff} [μm]	IC	G_{ds} [nA/V]	I_{D0} [μA]	V_E [V]	λ [$V/\mu m$]	Comment
pMOS	9.928	1	0.523	0.235	-448.547	44.855	moderate

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5 Noise

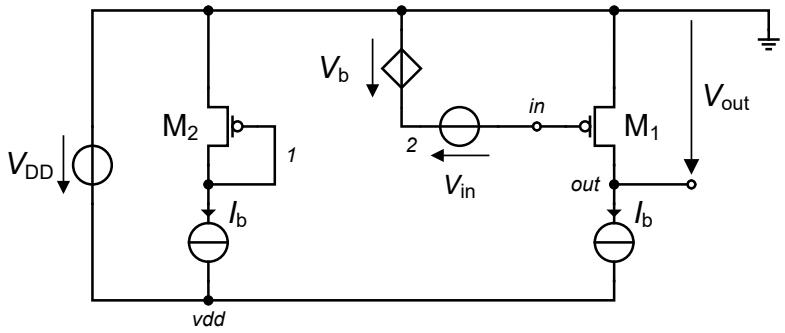


Figure 5.1: Schematic used for the noise simulation.

We now will check the noise model including the thermal and flicker noise. The simulations will be performed with the help of the schematic shown in Figure 5.1. Transistor M_1 is the active transistor and M_2, which is identical to M_1, is used to set the gate voltage of M_1 such that its current is equal to I_b.

5.1 Setting the bias conditions

Table 5.1: Transistor size and bias information.

Transistor	$W\ [\mu m]$	$L\ [\mu m]$	$I_D\ [nA]$	$I_{spec}\ [nA]$	IC	$V_G - V_{T0}\ [mV]$	$V_{DSsat}\ [mV]$
M1	10.00	10.00	175	175	1.0	15	116
M2	10.00	10.00	175	175	1.0	15	116

We choose an inversion coefficient $IC = 1$, which for the selected transistor size leads to a bias current $I_b = 0.175\ \mu A$.

Table 5.2: Transistor small-signal and thermal noise parameters.

Transistor	$G_{spec}\ [\mu A/V]$	$G_{ms}\ [\mu A/V]$	$G_m\ [\mu A/V]$	$G_{ds}\ [nA/V]$	γ_n
M1	6.776	4.188	3.207	0.394	0.736
M2	6.776	4.188	3.207	0.394	0.736

5.2 Extract operating point information

The node voltages are given in Table 5.3.

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Table 5.3: Node voltages.

Node	Voltage
vdd	-1.8
in	-0.482652
out	-0.482652
1	-0.482652
2	-0.482652
1	-0.482652

Table 5.4: Operating point information extracted from ngspice .op file for each transistor.

Transistor	I_D $[nA]$	I_{spec} $[nA]$	IC	n	V_{Dsat} $[mV]$
M1	175	171	1.027	1.31	156
M2	175	171	1.027	1.31	156

Table 5.5: Small-signal operating point information extracted from ngspice .op file for each transistor.

Transistor	G_m $[\mu A/V]$	G_{ms} $[\mu A/V]$	G_{ds} $[nA/V]$
M1	3.103	4.141	0.504
M2	3.103	4.141	0.504

Table 5.6: Noise operating point information extracted from ngspice .op file for each transistor.

Transistor	R_n $[k\Omega]$	$\sqrt{S_{ID,th}}$ $[nA/\sqrt{Hz}$	γ_n [-]	$\sqrt{S_{ID,fl}}$ at 1Hz $[nA/\sqrt{Hz}]$
M1	246.155	63.877	0.764	8940.59
M2	246.155	63.877	0.764	8940.59

5.3 Input-referred noise PSD

To compare the simulated noise to the theoretical noise PSD, we can simulate the circuit shown in . The noise at the output- and input-referred noise PSD will include both contributions of M_1 and M_2. SInce M_1 are identical and have the same bias current, we can simply divide the output- and inputreferred noise PSD by a factor 2. In ngspice to get the noise PSD we need to set `setplot noise1`. We can then export the noise data using `wrdata $inputdir/noise.nz.dat inoise_spectrum onoise_spectrum`. The simulated input-referred noise PSD is compared to the theoretical one in Figure 5.2.

We can extract the individual contributions of M_1 by the following instrauctions `save onoise_n1 onoise_n1_flicker onoise_n1_thermal` and saving the raw data with `write $inputdir/noise.nz.raw`. The resulting PSD are plotted in Figure 5.3.

We see that the simulation matches the theoretical estimation, particularly for the flicker noise. The simulated thermal noise is slightly larger than the theoretical estimation. This comes from the slightly smaller simulated transconductance compared to the theoretical value.

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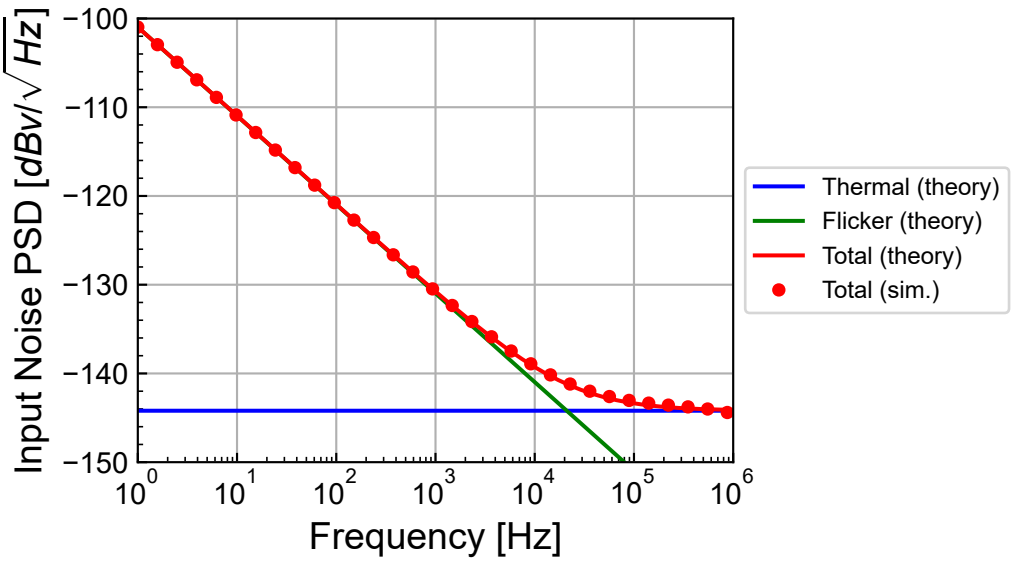


Figure 5.2: Simulated input-referred noise PSD of M_1 compared to theoretical estimation from `inoise_spectrum`.

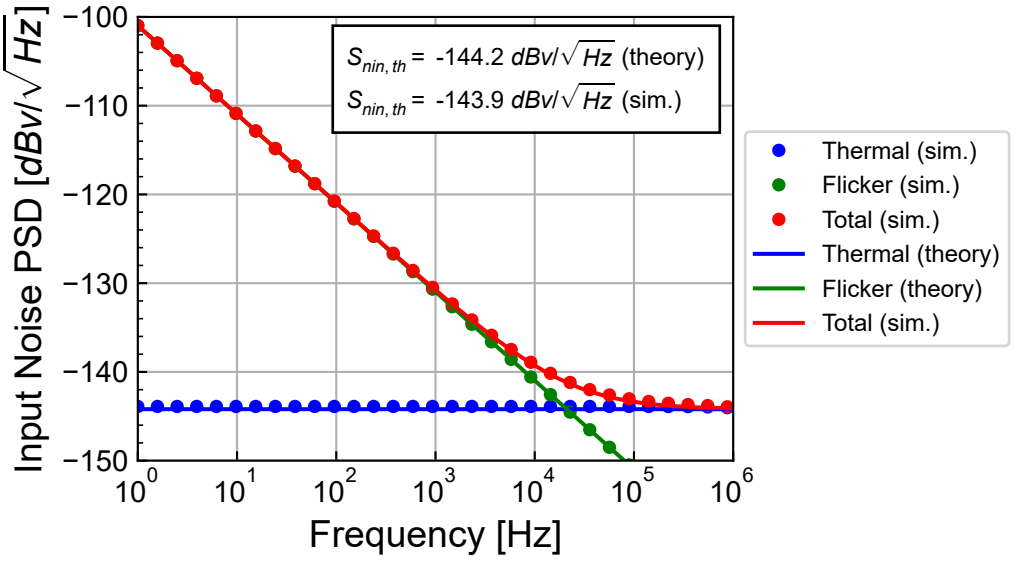


Figure 5.3: Simulated input-referred noise PSD of M_1 compared to theoretical estimation.

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6 Conclusion

This notebook presented different approaches to extract the sEKV parameters for a long-channel pMOS transistor for a generic 180nm bulk CMOS technology. The data was first obtained by simulations using the EKV 2.6 compact model.

The sEKV parameters were first extracted using a direct extraction methodology with $\lambda_c = 0$. This results in a good fit over all regions of operation except in very strong inversion because effect of mobility reduction due to the vertical field is not included in the sEKV model. Things can be improved by using the λ_c parameter to account for this effect although this is a very long-channel transistor.

Overall we have shown that sEKV can fit the large- and small-signal data of this is a long-channel pMOS transistor very well.

> **!** Important
>
> Although the extraction of the sEKV parameters using either direct extraction or by optimization give very good fit, we will use the parameters that are calculated from the EKV 2.6 compact model parameters adding L_{sat} and the CLM parameter λ. Indeed, this gives simulation results that are more consistent with the theoretical prediction, particularly for the inversion coefficient and the transconductances.

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